

A BUSINESS DATA MODEL CHATBOT

Objective: Build a RAG-based API to navigate the Microsoft Common Data Model (CDM).

Project Overview

At **reeeliance**, we help clients achieve "AI Readiness." This often involves moving from heterogeneous data to standardized, semantic models. The **Microsoft Common Data Model (CDM)** is a shared language for business entities (like Account, Lead, or Product).

Your Task: Build a Python-based FastAPI service that allows a user to query the CDM definition using natural language. The system should explain which entities exist, their attributes, and their relationships.

Technical Requirements

1. Data Ingestion:

- Source: Official Microsoft CDM GitHub Repository (<https://github.com/microsoft/CDM>)
- Scope: Focus on the **Banking Model** and common objects.
- Implementation: Parse these definitions and store them in a **Vector Database** of your choice.

2. RAG Pipeline:

- Implement a retrieval chain that answers questions like:
 - *"What are the core attributes of the 'Account' entity?"*
 - *"How does a 'Contact' relate to an 'Organization'?"*
- Ensure the LLM uses the CDM context to provide accurate, non-hallucinated answers.

3. Code Quality & DevOps:

- Use **FastAPI** for the backend.
- Include **Unit Tests** for your retrieval logic.
- Provide a **Dockerfile** for easy deployment.

4. Create a slide answering: How do I see myself as a Senior Data & AI Architect at reeeliance? Cover your core strengths, the experience you bring, and how you see your role evolving within the organization.

Deliverables (for the 3rd interview)

- A GitHub Repository link with your source code.
- A Live Demo of the API.
- A Technical Walkthrough (max. 4 slides) explaining your choice of embedding strategy and how you handled data relationships.